

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

5 Warfarin is used as an anticoagulant to treat some patients.

Patients respond differently to this drug. This makes it difficult for doctors to select a safe but effective dose of warfarin to give to each patient.

Factors that affect the response of a patient to warfarin include age, body mass, liver function and genetic makeup.

(a) Explain why a safe but effective dose of warfarin needs to be given to a patient.

(2)

.....

.....

.....

.....

.....

.....



- (b) There are two genes that code for proteins that affect warfarin and the blood clotting process.

Gene M:

- codes for enzyme M that breaks down warfarin into inactive molecules
- has many alleles and two of these code for much slower breakdown of warfarin.

Gene V:

- codes for enzyme V that converts vitamin K into an active form that triggers the blood clotting process
- one version of this gene results in a decreased production of the messenger RNA (mRNA) needed for the synthesis of enzyme V.

- (i) Describe the role of mRNA in the production of enzymes.

(2)

.....

.....

.....

.....

.....

.....



*(ii) Pharmacogenomics analyses the genetic profile of individuals.

This analysis is used to determine the appropriate doses of a drug for individual patients.

Discuss the possible use of pharmacogenomics in selecting appropriate doses of warfarin for individual patients.

Use the information in the question and your own knowledge of the blood clotting process and genetic screening to support your answer.

(6)

(Total for Question 5 = 10 marks)

